BUS ONBOARDING SYSTEM WITH PASSENGER PREFERENCE USING PYTHON

A Main Project Report Submitted in partial fulfillment of the requirements for award the degree of

MASTER OF COMPUTER APPLICATIONS

HINDUSTHAN COLLEGE OF ARTS & SCIENCE (AUTONOMOUS), Coimbatore.

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(Reg No: 23MCA038)

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DEPARTMENT OF COMPUTER APPLICATIONS (PG)

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APRIL -2025

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CERTIFICATE

This is certify that the project work entitled **"BUS ONBOARDING SYSTEM WITH PASSENGER PREFERENCE USING PYTHON"**, submitted to in partial fulfillment of the Requirements for the award of the Degree of Master of Computer Applications is a record of project work done by **KAVYA S (Reg no: 23MCA038)** under the supervision and guidance of **Dr. N. REVATHY,** Professor, Department of Computer Applications (PG), Hindustan College of Arts & Science, Coimbatore.

Guide **Director**

Submitted for Main Project Viva-Voice Examination held on.....

Internal Examiner **External Examiner**

DECLARATION

DECLARATION

I hereby declare that this project work entitled "**BUS ONBOARDING SYSTEM WITH PASSENGER PREFERENCE USING PYTHON" is** a record of original work done by me, under the supervision and guidance of **Dr.N. REVATHY, Professor**, Department of Computer Applications (PG), Hindusthan College of Arts & Science, Coimbatore.

Place: Coimbatore **Signature of the candidate**

Date: **(KAVYA S)**

ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be put incomplete without the mention of people who made it possible, whose constant guidance and encouragement crowed by efforts with success. I consider it my privilege to express my gratitude and respect to all those who guide and inspired me in completion of this main project

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I express my profound gratitude to our Head of the Department **Dr.A.V. SENTHIL KUMAR, MCA., M.Phil., P.G.D.C.A., Ph.D., Director** of PG and Research Department of Computer Applications, Hindusthan College of Arts & Science, Coimbatore for fruitful discussion, for constant support and encouragement and put me in right path, for valuable suggestions and timely help in each stage of my project

I take pleasure in thanking our project coordinator & guide **Dr.N. REVATHY, MCA., M.Phil., Ph.D., Professor,** Department of Computer Application (PG) for extending her support and guidance throughout the project. My sincere thanks to all faculty members of the, Department of Computer Applications (PG), for their supportive encouragement.

I owe my acknowledgement to one and all, who have directly or indirectly aided me in completing my project.

KAVYA S

ABSTRACT

Urban public transportation systems are fundamental to city life, facilitating daily commutes, economic activities, and environmental sustainability. However, existing systems primarily focus on operational efficiency rather than passenger satisfaction, leading to inefficiencies and underutilization. This paper presents a Python-based Bus Onboarding System with Passenger Preference to enhance both passenger experience and operational performance. The system leverages real-time data collection, route optimization, and dynamic scheduling to provide a personalized, efficient, and sustainable public transport solution. A unique feature of the system is a dynamic fare adjustment model that accounts for holidays and passenger preferences. Through improved passenger experience and optimized operations, the system reduces costs, enhances satisfaction, and minimizes environmental impact. The prototype implementation demonstrates its feasibility, scalability, and effectiveness in diverse urban settings.

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INTRODUCTION

CHAPTER-1
INTRODUCTION

The increasing urban population has led to rising demand for efficient public transportation, exacerbating challenges such as traffic congestion, resource strain, and environmental concerns. Traditional bus systems prioritize operational efficiency but often fail to address passenger needs, resulting in inadequate service experiences. This paper introduces the Bus Onboarding System with Passenger Preference, designed to provide a personalized transit experience, improve route efficiency, and reduce operational costs. A key innovation is its dynamic fare model, which adjusts fares based on proximity to holidays, incentivizing early bookings and managing demand surges effectively.

1.1 Overview

The Bus Onboarding System with Passenger Preference is a comprehensive solution designed to enhance the efficiency and convenience of public transportation. As urban populations continue to grow, traditional bus systems face challenges such as congestion, resource mismanagement, and poor passenger satisfaction. By leveraging real-time data analytics and intelligent scheduling, this system provides an innovative and passenger-centric approach to modernizing transportation networks.

1.1.1 Key Features:

- **Real-Time Data Analytics:** Provides accurate insights for optimal scheduling.
- **Dynamic Fare Model:** Adjusts pricing based on demand, booking time, and special events.
- **User-Friendly Interface:** Enables seamless ticketing and tracking via web and mobile applications.
- **Scalability:** Adaptable to various urban environments with flexible expansion capabilities.

1.2. Module Description

The system is divided into several functional modules, each playing a vital role in its overall performance and reliability:

1.2.1 Passenger Preference-Based Scheduling:

- Collects and analyzes user preferences for travel times, seating arrangements, and frequent routes.
- Optimizes bus schedules dynamically.

1.2.2 Real-Time Route Optimization:

- Uses predictive algorithms such as Dijkstra's and A* to determine the most efficient routes.
- Integrates with real-time traffic data sources to avoid congestion.

1.2.3 Dynamic Fare Adjustment:

- Implements a flexible pricing model that adjusts fares based on booking time, demand levels, and special event periods.
- Encourages early bookings with lower fares and manages peak-time loads efficiently.

1.2.4 Integrated Payment and Ticketing:

- Supports multiple payment methods, including credit cards, digital wallets, and NFC-based transactions.
- Ensures a secure and seamless ticketing experience with digital passes and QR codes.

1.2.5 Data-Driven Decision Making:

- Provides an analytics dashboard for transportation authorities to monitor passenger trends, optimize routes, and improve service delivery.
- Collects and processes feedback to refine future operations.

SYSTEM STUDY

CHAPTER-2
SYSTEM STUDY

2.1 Existing System

Traditional bus onboarding systems primarily focus on static scheduling and fixed-route services. These systems lack flexibility, often resulting in:

- Inconsistent service quality due to fluctuating passenger demand.
- Inefficient route planning leading to overcrowding or underutilized buses.
- Fixed pricing structures that do not adapt to demand variations.
- Minimal integration of real-time passenger feedback.

2.2 Proposed System

The proposed Bus Onboarding System integrates real-time data analytics and dynamic fare pricing to improve passenger experience and optimize operational efficiency. The key features include: Passenger Preference-Based Scheduling: Personalized route and seating options based on user history.

- Real-Time Route Optimization: Use of predictive algorithms to adjust routes dynamically.
- Dynamic Fare Adjustment: Pricing strategies based on booking times and demand fluctuations.
- Integrated Payment and Ticketing: Seamless digital transactions via multiple payment methods.
- Data-Driven Decision Making: Dashboard analytics for improved transportation planning.

SYSTEM SPECIFICATION

CHAPTER-3
SYSTEM SPECIFICATION

3.1. Programming Language

- **Python**: The main backend language for server-side processing, database interaction, and business logic implementation.
- **JavaScript**: For handling frontend interactivity (e.g., updating bus schedules dynamically).

3.1.1. Web Framework

Flask or Django: Used to build the web-based application and manage backend logic.

- **Flask** is lightweight and easy to implement.
- **Django** offers built-in authentication, an admin panel, and robust database handling.

3.1.2. Database Management System (DBMS)

- **SQLite / PostgreSQL / MySQL**
  - **SQLite**: For lightweight applications.
  - **PostgreSQL or MySQL**: If deploying at scale.
  - **Tables to Store Data**:
    - Passenger details (ID, Name, Contact)
    - Booking details (Source, Destination, Fare, Time)
    - Bus schedules (Route ID, Stops, Timings)
    - Payment transactions

3.1.3 Frontend Technologies

HTML, CSS, JavaScript (with Bootstrap)

- **Bootstrap**: Ensures a responsive UI.
- **JavaScript & AJAX**: Allows real-time updates for bus schedules and availability.

3.1.4 APIs for Route Optimization (Without AI/ML)

- **Google Maps API / OpenStreetMap API**: To fetch traffic data and find optimal routes.
- **Geolocation API**: To determine the user's location and nearby bus stops.

3.1.5 Payment Gateway

- **Stripe / PayPal / Razosssrpay** for handling online payments securely.

3.1.6 Data Visualization

- **Matplotlib / Seaborn / Plotly**: Used to generate analytics like passenger booking trends.

3.2. Hardware Specification

1. **Processor**: Intel Core i5 or higher
2. **RAM**: Minimum 8GB (recommended 16GB for better performance)
3. **Storage**: At least 256GB SSD
4. **Network**: Stable internet connection for API access and cloud integration
5. **Server (Optional for Deployment)**: Cloud services like AWS, Google Cloud, or a dedicated server

3.3. Software Description

1. **Operating System**: Windows / Linux / macOS
2. **Database Software**: PostgreSQL / SQLite
3. **Web Server**: Apache / Nginx (if hosted on a server)
4. **Python Libraries**: Flask/Django, Pandas, NumPy, Matplotlib
5. **Development Tools**: VS Code / PyCharm / Jupiter Notebook

SYSTEM DESIGN

CHAPTER-4
SYSTEM DESIGN

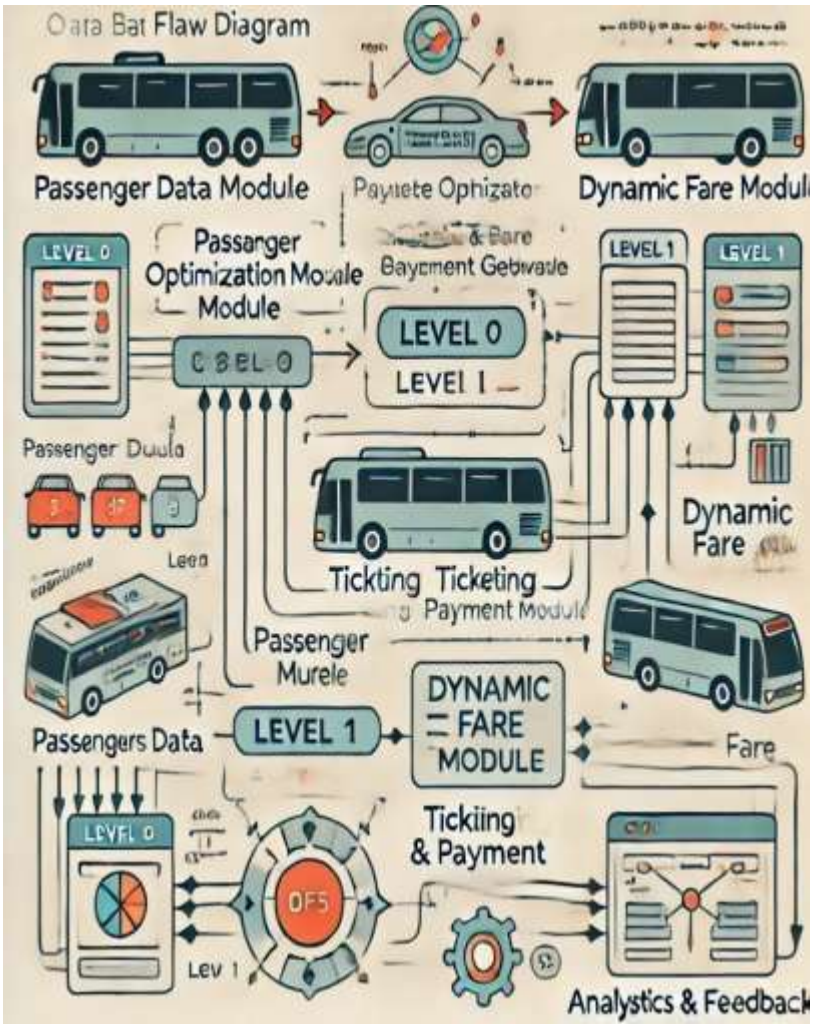
4.1. Data Flow Diagram (DFD)

Level 0 - Context Diagram: The Bus Onboarding System interacts with:

- **Passengers** (Input preferences, bookings, payments)
- **Bus Operators** (Manage schedules, routes, pricing)
- **Payment Gateway** (Processes transactions)
- **Traffic API** (Provides real-time traffic updates)

Level 1 DFD:

- **Passenger Data Module**: Captures user inputs (preferences, bookings, payments)
- **Route Optimization Module**: Processes route data and schedules
- **Dynamic Fare Module**: Adjusts fare based on booking time and demand
- **Ticketing & Payment Module**: Generates tickets and verifies transactions
- **Analytics & Feedback Module**: Analyzes service quality and user satisfaction

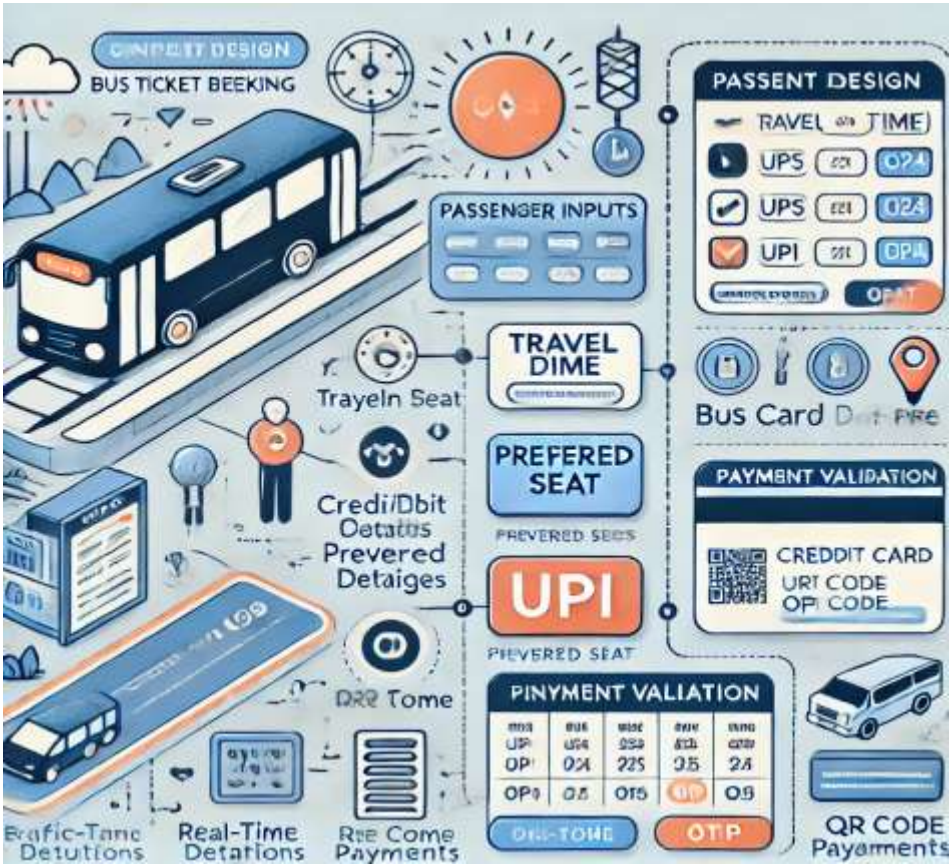


4.2 Input Design

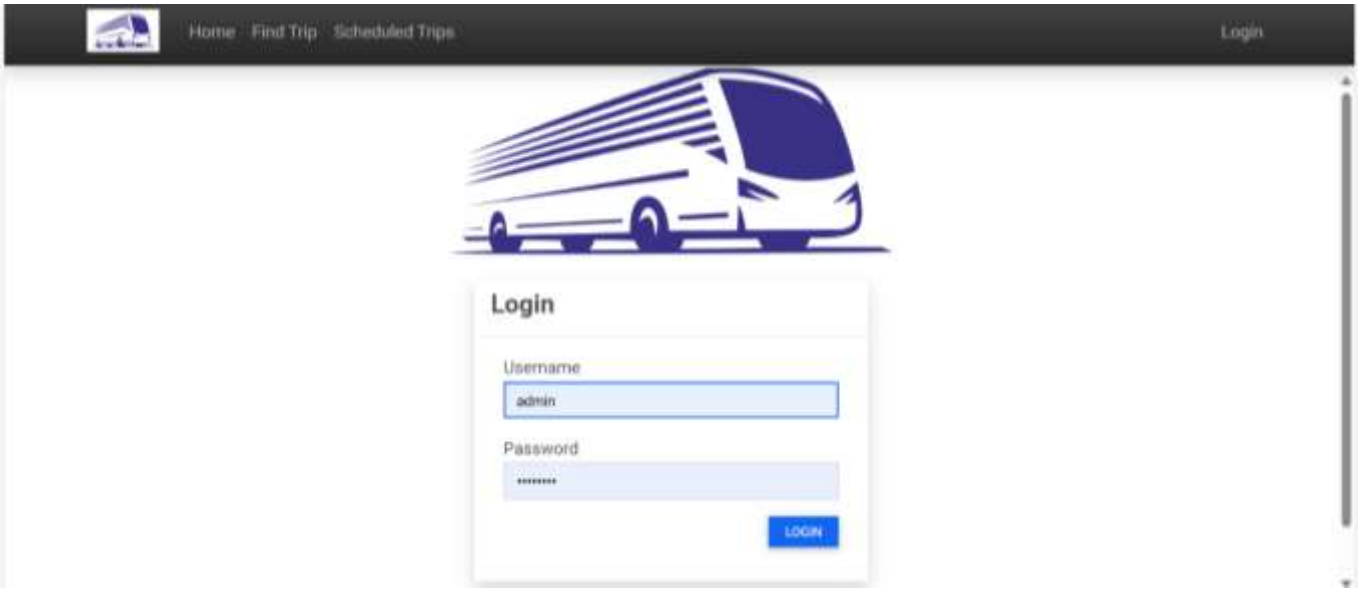
- **Passenger Inputs**: Travel date, time, preferred seat, route selection
- **Bus Operator Inputs**: Bus schedules, route changes, fare adjustments
- **Payment Inputs**: Credit/debit card details, UPI, QR code payments
- **Real-Time Inputs**: Traffic data, weather conditions, demand trends

Input Validation Techniques:

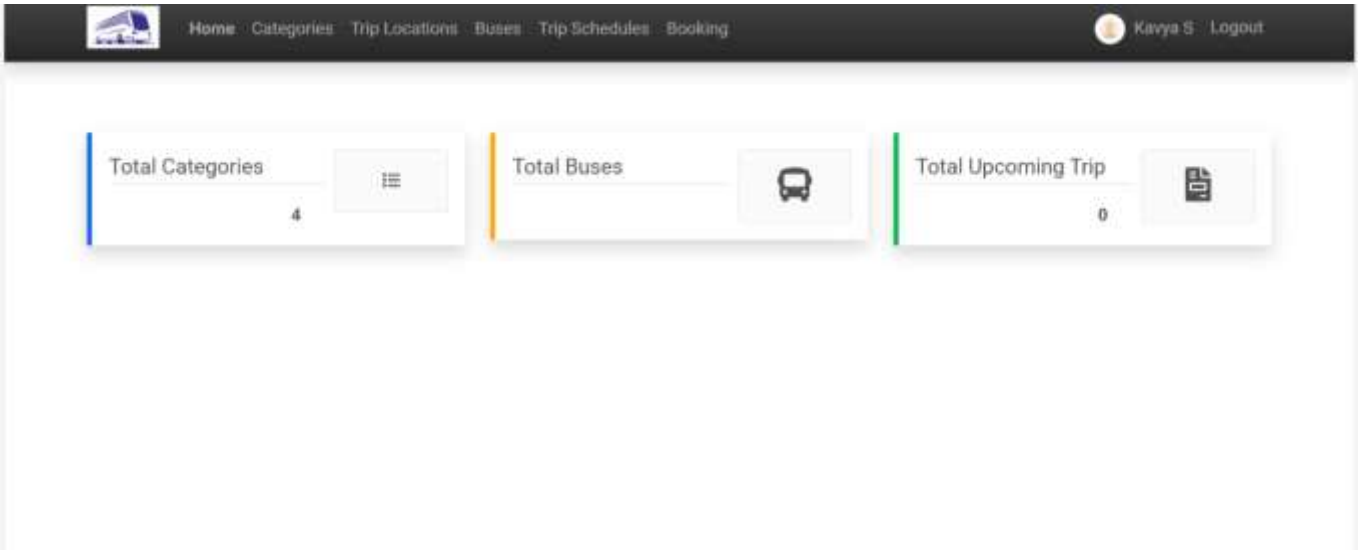
- Dropdown selections for route preferences
- Date pickers for travel dates
- OTP verification for secure logins and payments



ADMIN PANEL



HOME PAGE



4.3 System Design

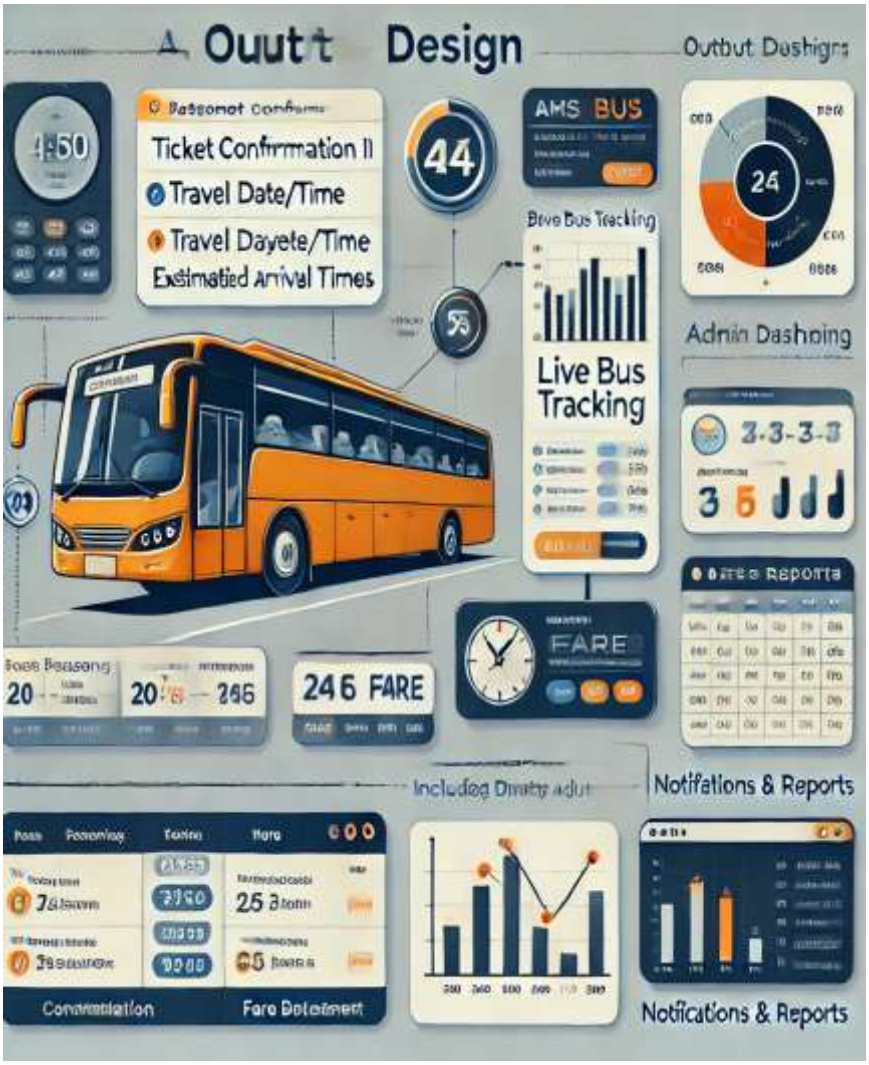
- Ticket confirmation details (Booking ID, travel date/time, fare)
- Live bus tracking and estimated arrival times
- Fare breakdown including dynamic adjustments

Admin Dashboard Outputs:

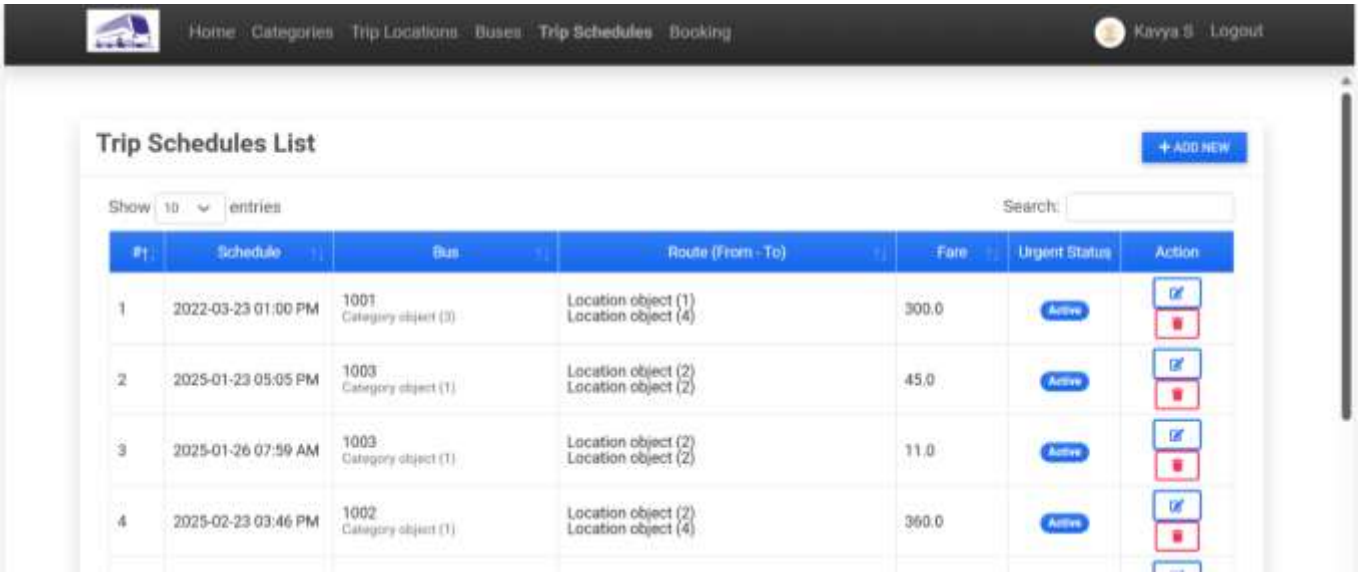
- Real-time bus utilization and passenger booking trends
- Route efficiency metrics
- System-generated fare reports

Notifications & Reports:

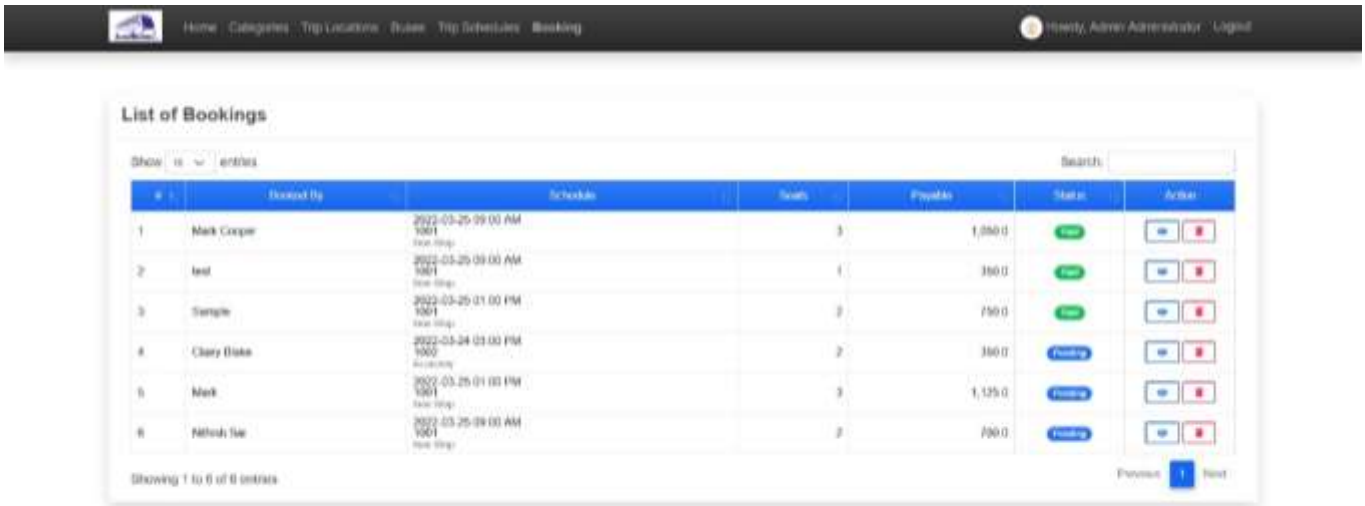
- Email/SMS confirmations for bookings
- Alerts for route changes or delays
- Monthly passenger trend analysis



TRIP SCHEDULE LIST



LIST OF BOOKINGS



SYSTEM TESTING

CHAPTER-5
SYSTEM TESTING

5.1 Unit Testing

. Unit testing ensures that individual components of your system function correctly. In your case, key modules to test would be:

5.1.1 Passenger Data Collection Module

Test if passenger preferences (route, seat selection, time) are correctly stored and retrieved from the database.

5.1.2 Route Optimization Algorithm

Verify that the algorithm (Dijkstra's or A*) returns the optimal route given real-time data.

5.1.3 Dynamic Scheduling

Check if schedule adjustments work correctly based on passenger demand and traffic data.

5.1.4 Ticketing and Payment Gateway

Ensure payment processing via Stripe/PayPal works without errors.

5.1.5 Fare Calculation (Dynamic Fare Model)

Validate that fares change correctly based on the proximity to holidays.

5.1.6 Analytics Dashboard

Test data visualization components to confirm accurate data representation.

Test Case No	Test Case	Description	Expected Result	Observed Result	Status
1	Enter username and password	Check valid username and password	The username and password should be accepted	The username and password entered correctly	Pass

5.2 Validation testing

Validation testing ensures that the system meets user requirements and functions as expected. It involves testing all modules based on user input to verify that the system accepts valid input and produces the expected output. A successful validation test confirms that the system is reliable and meets the specified requirements.

Test Case No	Test Case	Description	Expected Result	Observed Result	Status
1	Testing all system modules	Tests all modules based on user input	System should accept valid input and provide expected output	System produces valid and reliable output	Pass

5.3 System testing

Integration testing verifies the combined functionality of all modules after integration. It ensures that individual components work together seamlessly, producing valid and reliable outputs when tested with real user inputs. A successful integration test confirms that the system functions correctly as a whole.

Test Case No	Test Case	Description	Expected Result	Observed Result	Status
1	Testing all modules in system	Ensure all modules execute correctly	System should accept and execute all modules successfully	System accepts all modules and produces expected result	Pass

5.4 Integration testing

System testing evaluates the complete system to ensure all modules function correctly together. It verifies that the system meets functional and performance requirements, processes inputs correctly, and produces expected results. A successful system test confirms the system's overall reliability and effectiveness.

Test Case No	**Test Case**	**Description**	**Expected Result**	**Observed Result**	**Status**
1	Verify combined functionality after integration	Test all modules with user inputs to ensure smooth operation	System should produce a valid, reliable output after integration	System produces expected output from all modules	Pass

CONCLUSION

CHAPTER-6
CONCLUSION

This advanced system marks a transformative leap in the evolution of urban public transportation, seamlessly blending efficiency with passenger-centric innovation. By leveraging cutting-edge technologies, it not only enhances operational effectiveness but also redefines the commuter experience. A key highlight of this system is its **dynamic fare model**, which intelligently adjusts pricing based on real-time demand patterns. This ensures a more balanced distribution of passengers, prevents overcrowding, and optimizes resource utilization, ultimately leading to a more sustainable and efficient transit network.Additionally, **real-time route optimization** stands at the core of this solution, dynamically adjusting routes based on live traffic conditions, passenger demand, and unforeseen disruptions. This feature guarantees improved service reliability, reduced travel times, and enhanced adaptability, ensuring that commuters experience minimal delays and maximum convenience. Looking ahead, future developments will focus on **scaling the system for larger metropolitan areas**, incorporating multi-modal transit options, and seamlessly integrating with broader **smart city ecosystems**. By connecting with AI-powered traffic management, IoT-enabled infrastructure, and mobility-as-a-service (MaaS) platforms, this system will play a crucial role in shaping the next generation of urban mobility. With its holistic approach to modernization, this solution paves the way for **a smarter, greener, and more passenger-friendly future**, redefining how cities move and thrive in the digital age.

FUTURE ENHANCEMENT

CHAPTER-7
FUTURE ENHANCEMENTS

7.1. Integration with Multi-Modal Transport Systems

- **Description**: To improve urban mobility, the system could be extended to integrate multiple modes of transport such as trains, trams, subways, taxis, and shared mobility services
- **Benefit**: Offers passengers more flexibility, convenience, and options, reducing the need to rely on a single transport mode and enhancing overall system efficiency.
- **Example**: A user could plan a route that combines a bus ride to the train station, followed by a subway ride, with the system providing integrated fare management and real-time updates.

7.2. Personalized Travel Recommendations and Dynamic Route Adjustments

- **Description**: Using advanced machine learning algorithms, the system could predict and personalize travel routes for users based on their historical preferences, real-time traffic conditions, and live updates about congestion or incidents.
- **Benefit**: Enhances passenger satisfaction by offering more personalized and flexible travel options. The system can continuously adapt to individual needs, helping users save time and effort by suggesting the most optimal routes.
- **Example**: If a user frequently takes a bus from their office to a gym, the system could prioritize this route and suggest alternatives based on current traffic conditions.
-

7.3. Dynamic Fare System Based on Demand and Proximity to Holidays

- **Description**: As holidays approach, the bus fare will be a fixed amount up to 10 days before the holiday. However, closer to the holiday, the fare will increase for passengers who are rushing to catch the bus. This dynamic pricing strategy will balance the demand during high-traffic periods, ensuring that passengers who plan ahead benefit from lower fares, while those requiring immediate travel will pay a higher fare.
- **Benefit**: Promotes early bookings and better resource management while ensuring that the system can handle increased demand during peak periods such as holidays.
- **Example**: If a holiday is on the 15th of the month, the fare will remain fixed from the 5th to the 10th, but after the 10th, passengers will be charged progressively higher fares depending on how close they are to the holiday date.

7.4. AI-Powered Traffic Management and Real-Time Schedule Adjustments

- **Description**: By incorporating artificial intelligence, the system can predict traffic congestion and automatically adjust bus schedules to minimize delays. AI can process data from traffic cameras, sensors, and historical patterns to offer real-time updates and reroute buses to avoid delays caused by accidents or road closures.

- **Benefit**: Improves the efficiency of the system by minimizing delays and disruptions. Passengers will experience fewer delays, and operational costs can be reduced by optimizing bus schedules based on real-time conditions.

- **Example**: If an accident occurs on a bus route, the system can dynamically adjust the schedule, notify passengers of the change, and suggest alternative routes, all while ensuring the bus fleet is optimized for demand.

7.5. Seamless Integration with Smart City Infrastructure

- **Description**: The system could be integrated with existing smart city infrastructure, such as traffic lights and pedestrian walkways, enabling real-time traffic signal adjustments based on bus schedules and passenger needs. This integration can streamline the overall transportation network, reducing travel time and improving traffic flow.

- **Benefit**: Enhances the efficiency of public transportation by ensuring that buses are prioritized at traffic signals and that the entire transportation system is optimized in real-time.

- **Example**: Buses equipped with GPS can communicate with traffic lights, ensuring that they turn green as buses approach, minimizing delays and improving punctuality.

7.6. Voice-Activated Services and Accessibility Features

- **Description**: Integrating voice recognition technology would allow passengers to interact with the system using natural language commands, such as checking bus schedules, booking tickets, or providing feedback.

- **Benefit**: Makes the system more user-friendly and inclusive, enhancing the travel experience for people with disabilities or those unfamiliar with technology.

- **Example**: A passenger with limited vision could simply ask, "What is the next bus to my destination?" and receive voice-based updates on the schedule and bus availability.

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BIBLIOGRAPHY

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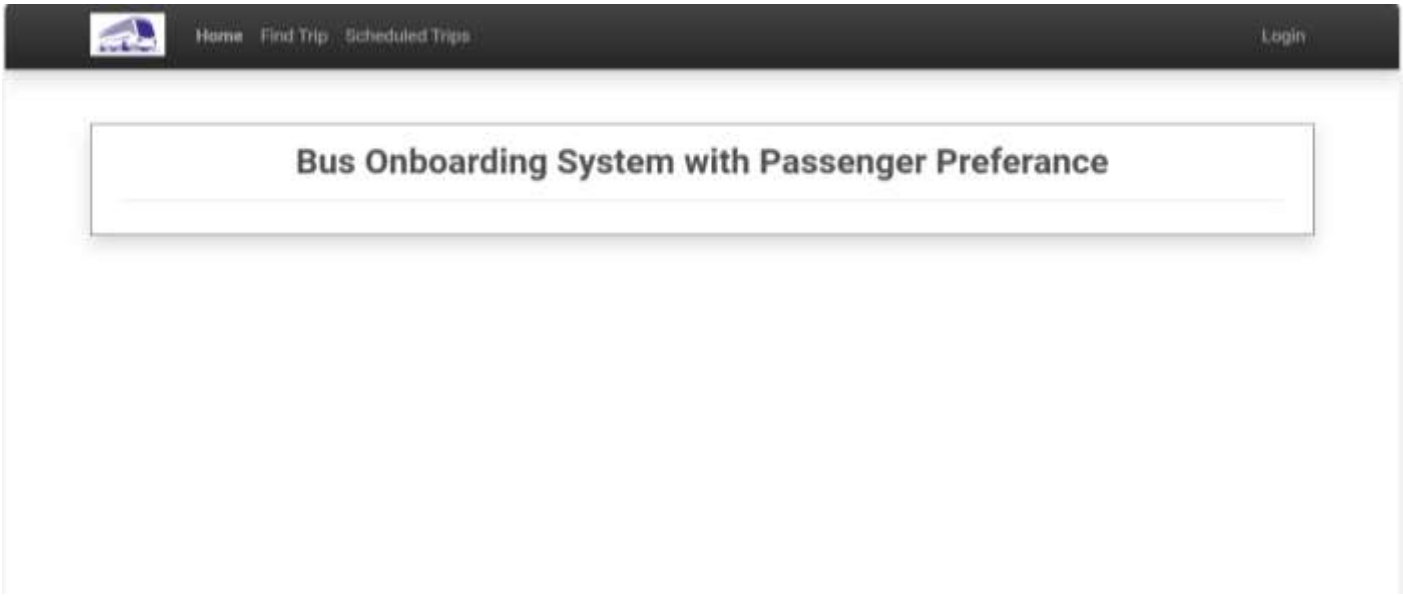
APPENDIX

A. SCREENSHOTS

ADMIN PANEL



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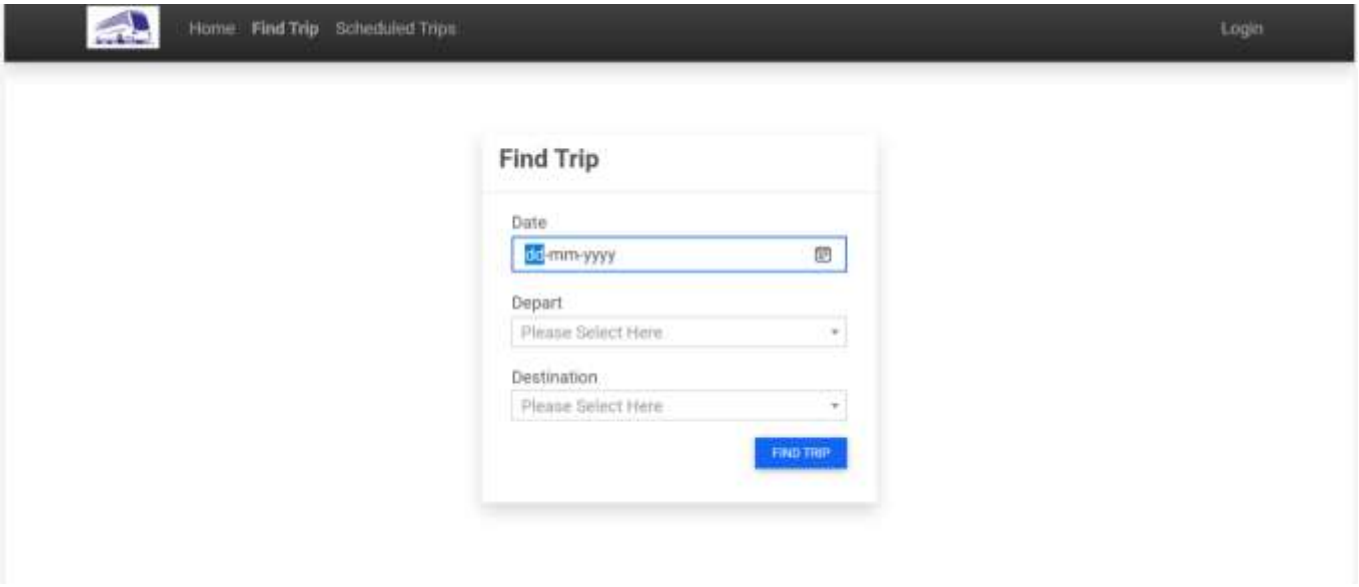

SCHEDULED TRIP

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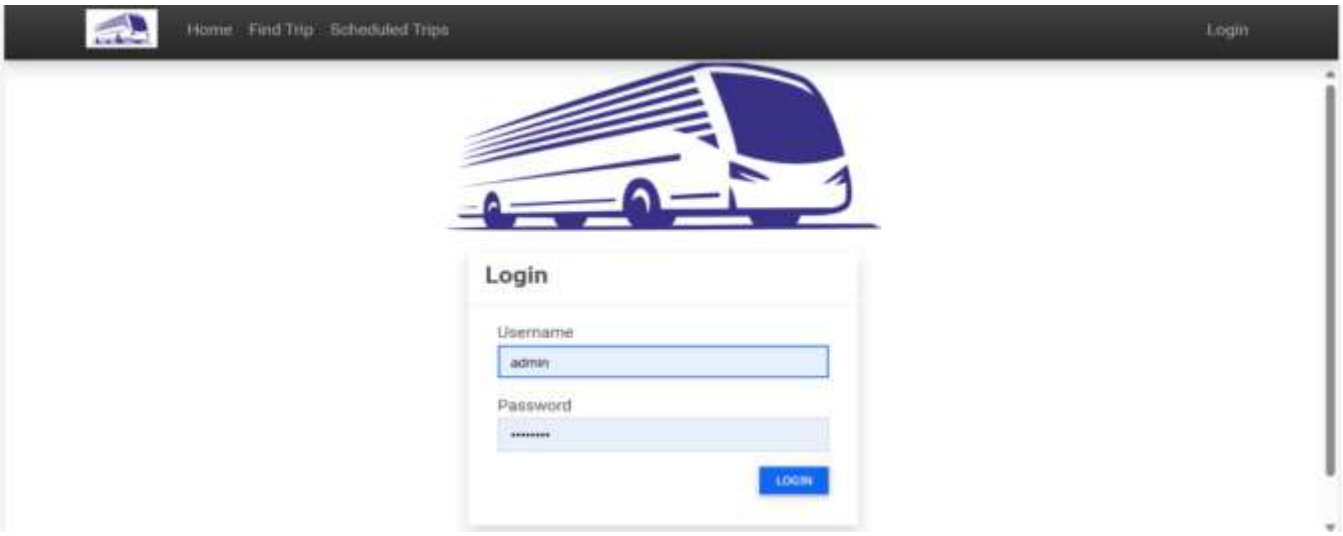

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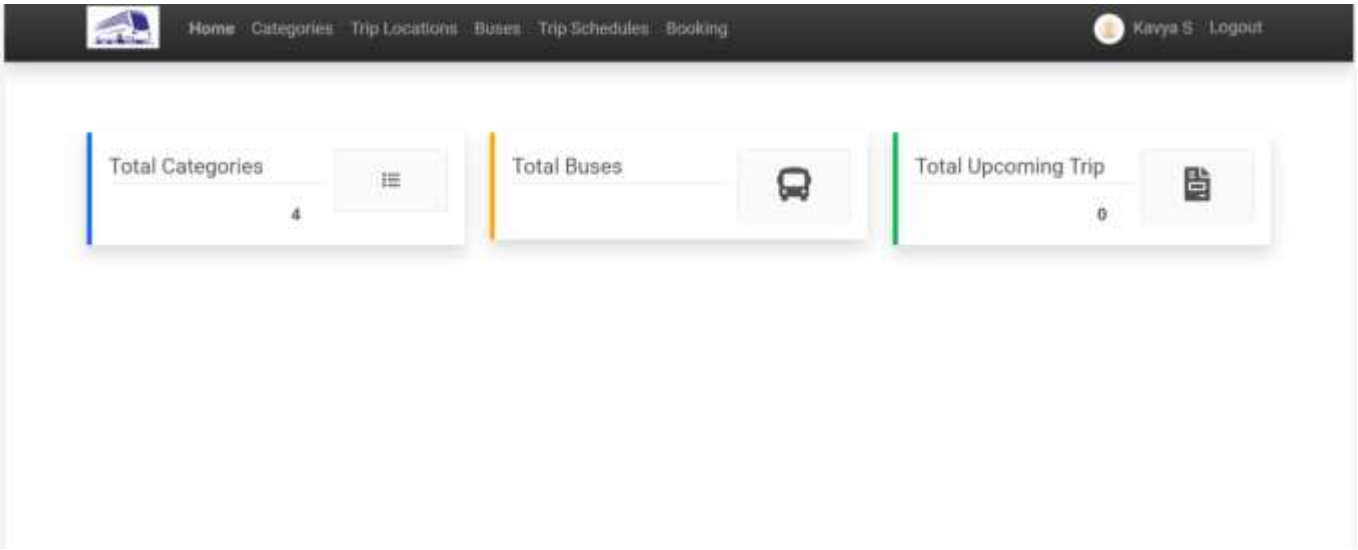
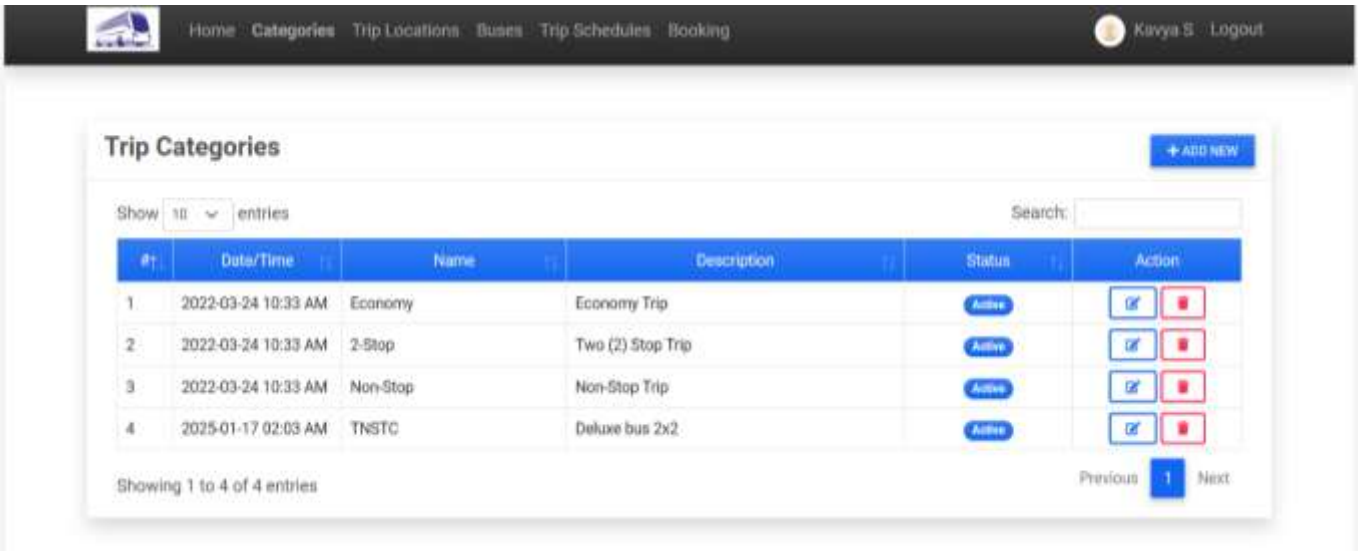
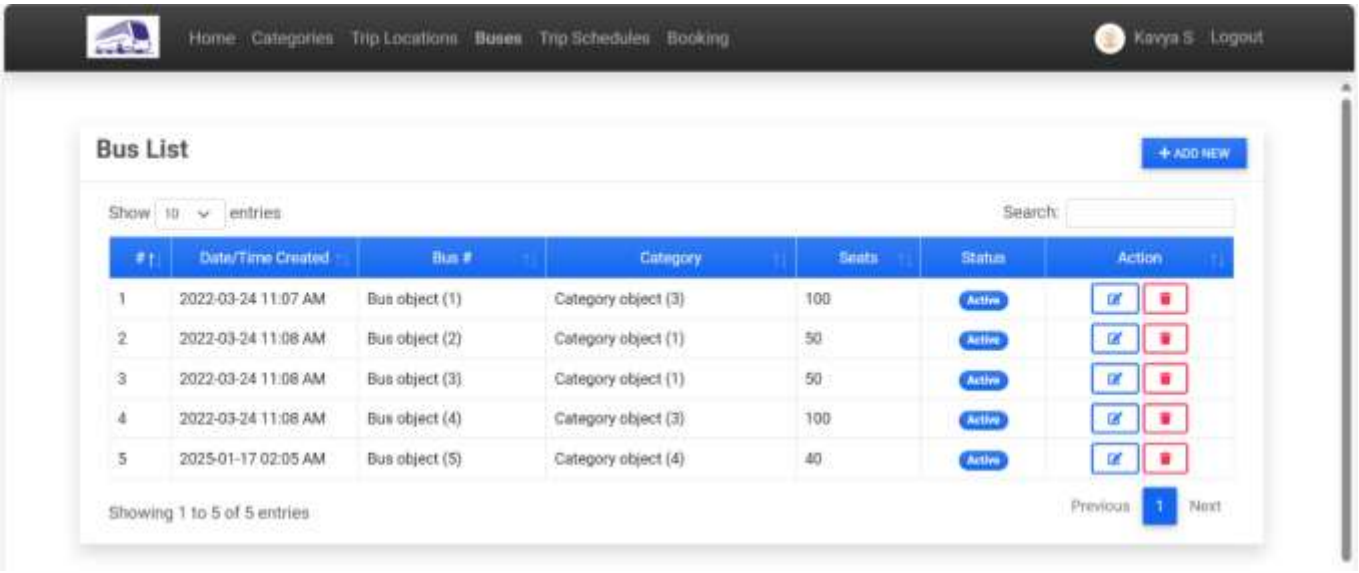


TRIP CATEGORIES



BUS LIST



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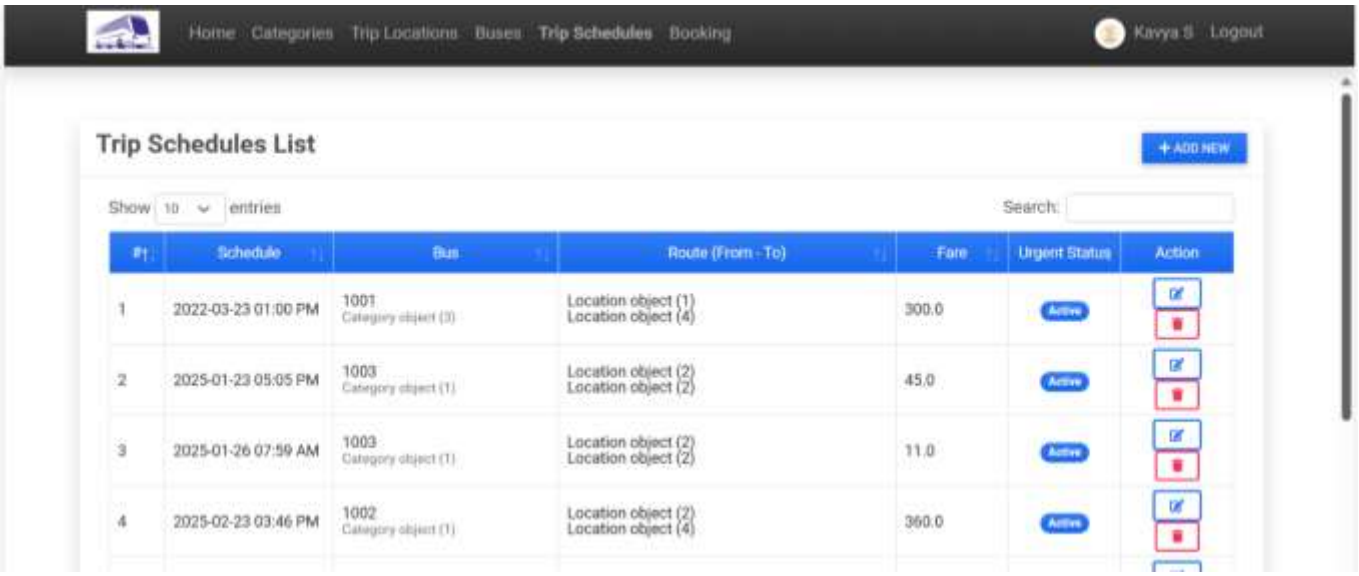
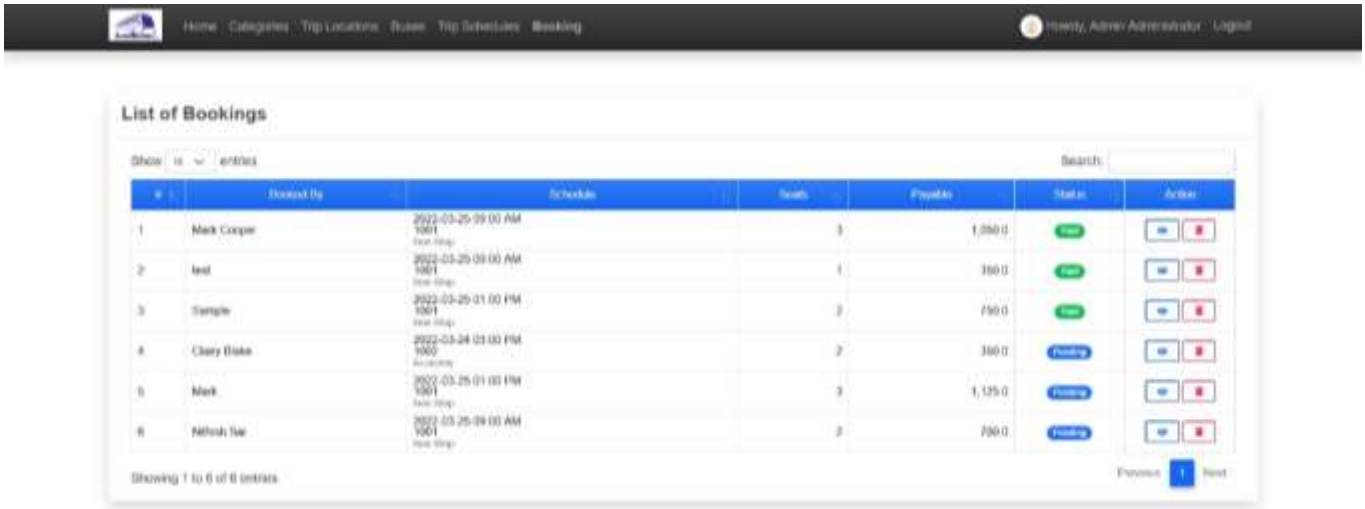


LIST OF BOOKING



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in the 6th International Conference on Computer Applications and Information Technology
organised by Department of Computer Applications (PG) in
Hindusthan College of Arts & Science, Coimbatore
on 20th February 2025.

Dr. N. Revathy, Mrs. V. Latha Sivasankari	Dr. A. V. Senthil Kumar	Dr. A. Ponnusamy
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BUS ONBOARDING SYSTEM WITH PASSENGER PREFERENCE

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Abstract

Urban public transportation systems are integral to city life, facilitating daily commutes, boosting economic activity, and reducing environmental footprints. However, existing systems often emphasize operational efficiency at the expense of passenger needs, leading to suboptimal satisfaction and underutilization. This paper introduces a Python-based Bus Onboarding System with Passenger Preference, aimed at enhancing both passenger experience and operational efficiency. The system integrates real-time data collection, machine learning algorithms, route optimization, and dynamic scheduling to offer a personalized, efficient, and sustainable public transport solution. A unique feature of this system is the dynamic fare adjustment model, which incorporates holidays and passenger preferences. By addressing passenger preferences and operational challenges, the system improves satisfaction, reduces costs, and minimizes environmental impact. The prototype implementation demonstrates its feasibility, scalability, and effectiveness across various urban environments.

Keywords - Smart Transportation, Passenger-Centric Systems, Public Transport Optimization, Python Applications, Route and Schedule Management, Machine Learning in Transit, Dynamic Fare System

1. Introduction

Urban transportation faces rising demand due to growing populations, resulting in traffic congestion, resource strain, and environmental challenges. Traditional bus systems typically focus on operational efficiency, leaving passengers with limited flexibility and poor service experience. By utilizing machine learning and real-time data analytics, public transport systems can better respond to the evolving needs of passengers while optimizing resources. This paper introduces the Bus Onboarding System with Passenger Preference, which aims to provide a personalized transit experience, improve route efficiency, and reduce operational costs

A key innovation in this system is the dynamic fare model, which incorporates the proximity to holidays as a factor in fare calculation. Passengers booking far in advance will benefit from lower fares, while those booking closer to the holiday will be subject to higher fares, reflecting the increased demand and urgency.

Key Goals:

1. Improve passenger satisfaction through personalized services.
2. Optimize bus schedules and routes to align with real-time demand.
3. Reduce costs and environmental impact via efficient resource allocation.
4. Incorporate dynamic fare pricing based on proximity to holidays to optimize demand.

2.Literature Review

Urban public transportation systems are essential to the functioning of modern cities, providing a reliable and cost- effective means of travel for millions of people every day. However, despite their importance, many traditional systems still focus primarily on operational efficiency rather than passenger satisfaction. This literature review explores the development and application of passenger-centric transport systems, highlighting recent advancements in machine learning, data analysis, real-time optimization, and dynamic fare systems that contribute to improving public transportation. It also covers previous studies that focus on personalized transport systems, the integration of technology in public transit, and the importance of passenger preferences in shaping future transit solutions.

2.1. Personalized Public Transportation Systems

The concept of personalized transportation has gained significant attention in recent years due to the increasing demand for customized services in public transport. A study by Zhang et al. (2020) emphasizes the importance of personalizing public transport services to improve passenger satisfaction and optimize resource utilization. According to their research, incorporating user preferences such as preferred travel times, routes, and seating arrangements can significantly improve the user experience. This personalization leads to increased public transport usage and reduced congestion by aligning the system more closely with individual needs.

Similarly, Fan et al. (2019) argue that personalized services, such as flexible routes, real-time scheduling, and demand- based allocation of resources, can enhance the efficiency of urban transport systems. They highlight how modern technologies, particularly mobile apps and

machine learning, can be used to gather and analyse passenger data to deliver personalized recommendations and adjust service offerings in real time.

2.2 Machine Learning and Data Analytics in Public Transit

Machine learning and data analytics have been pivotal in optimizing public transportation systems. Liu et al. (2018) describe how machine learning algorithms can predict passenger demand, optimize routes, and manage schedules dynamically. The application of machine learning in transportation systems allows for predictive analytics, which can reduce overcrowding, prevent delays, and improve operational efficiency. The integration of real-time data, including traffic information, passenger load, and weather conditions, further enhances the ability to optimize routes and schedules. According to a study by Chen et al. (2020), integrating machine learning techniques with traditional transportation management systems enables predictive routing and scheduling, which can reduce operational costs and increase the quality of service. Additionally, their research found that incorporating real-time data into decision-making processes helped minimize delays and provided a more responsive transportation service for passengers.

2.3 Route Optimization Algorithms

Route optimization is a fundamental aspect of improving public transport systems. Algorithms such as Dijkstra's Algorithm and A* are commonly used in transport systems to identify the shortest path between two points. Recent research, including work by Smith et al. (2017), has shown that the use of advanced optimization algorithms combined with real-time data can dramatically improve the efficiency of public transport systems. These algorithms consider factors such as traffic conditions, road closures, and passenger demand to determine the most efficient routes, minimizing delays and reducing fuel consumption.

Another study by Kotsiopoulos et al. (2018) explored how real-time traffic updates via APIs (e.g., Google Maps or OpenStreetMap) can be integrated into public transportation systems to dynamically adjust bus routes. The study found that dynamic rerouting significantly improved the overall travel time and satisfaction of passengers during peak hours.Dynamic Fare Systems and Pricing Strategies

Dynamic fare systems are an essential component of modern transport systems, particularly in managing demand and ensuring fairness. Several studies have explored the implementation of surge pricing and dynamic fare models based on demand, time of day, and proximity to holidays. For example, a study by Jiang et al. (2021) analysed the impact of

dynamic pricing on public transport systems and found that fare adjustments based on demand and time significantly improved system efficiency. Their research showed that passengers were more likely to adjust their travel habits if they were incentivized by lower prices during off-peak hours.

In terms of holiday-specific fare adjustments, a study by Zhou et al. (2019) found that transport systems that adjusted fares in anticipation of higher demand, such as during holidays, could reduce overcrowding and ensure a more balanced distribution of passengers. This strategy of applying a fixed fare up to a certain date before holidays and higher fares closer to the holiday could help optimize bus load management and ensure that buses run more efficiently during periods of high demand.Integration of Real-Time Scheduling and Traffic Management

Real-time scheduling is a growing area of interest in the transportation field. In a study by Gao et al. (2017), the authors discuss the importance of dynamic scheduling in response to fluctuating passenger demand and traffic conditions. The integration of GPS data and traffic management systems allows buses to adjust their schedules based on real-time conditions, such as accidents or congestion. This results in more accurate travel times and improved passenger satisfaction.

Sustainability in transportation is becoming an increasingly important concern, with many cities aiming to reduce their carbon emissions and improve the environmental footprint of their public transport systems. Studies such as that by Bock et al. (2020) argue for the integration of electric and hybrid buses into public transportation fleets. The transition to sustainable transportation not only reduces the environmental impact of the transport sector but also encourages passengers to adopt greener commuting habits.

3.System Design and Methodology

3.1 Architecture Overview

The **Bus Onboarding System** incorporates the following modules, which collectively enhance the passenger experience, improve operational efficiency, and enable dynamic fare pricing:

3.1.1. Passenger Data Collection

- Captures passenger preferences, including preferred travel times, seat selection, and route choices, via a **mobile app** or **web portal**.

- The data is stored in a **centralized database** for further analysis.

3.1.2. Route Optimization

- Uses **Dijkstra's Algorithm** and **A*** for determining the most efficient routes.
- Integrates real-time traffic updates through APIs like **Google Mapsor OpenStreetMap** for better route predictions.

3.1.3 Dynamic Scheduling

- Utilizes **predictive analytics** to adjust bus schedules in real-time, considering demand trends, weather conditions, and road congestion.
- The system adapts to the booking volume and adjusts buses accordingly.

3.1.4. Ticketing and Payment Gateway

- Supports **digital payment platforms** (such as **mobile wallets**, **QR codes**, and **NFC cards**) for streamlined ticket booking.
- Ensures a seamless ticket issuance process.

3.1.5. Analytics Dashboard

- Provides real-time **visual analytics** on metrics such as **passenger flow**, **route efficiency**, **bus utilization**, and **satisfaction scores**.

3.2. Components of the System

a. **Passenger Preference Module**

i. Captures data on preferred travel times, seat preferences, and onboard services through a user-friendly interface.

ii. Uses clustering algorithms to predict future preferences and recommend personalized services.

b. **Route Optimization Algorithm**

i. Uses machine learning and graph-based algorithms to analyze traffic conditions, passenger demand density, and fuel efficiency to recommend the best routes.

3. Machine Learning Frameworks: Scikit-learn, TensorFlow
4. Web Development Tools: Flask/Django
5. Database Management: SQLite, PostgreSQL
6. Visualization Tools: Matplotlib, Seaborn, Plotly
7. Payment Gateway: Stripe/PayPal

5. Workflow Diagram

- **Input Phase:** Passengers enter their preferences (e.g., time, seat, route) via the app or web portal.
- **Data Analysis:** The system analyzes passenger data using clustering and predictive models to optimize service offerings.
- **Route & Schedule Optimization:** The system calculates optimal routes and adjusts schedules based on real-time data.
- **Fare Calculation**: If a holiday is near, the fare is dynamically adjusted based on the proximity to the holiday.
- **Ticketing and Payment:** Passengers pay using preferred payment methods, and digital tickets are issued for a seamless onboarding experience.

6. Output Phase:

Passengers receive real-time updates, including bus locations, seat availability, and fare information. Applications

The Bus Onboarding System with Passenger Preference offers several significant benefits:

- Personalized Commuting: Passengers receive a tailored experience based on their travel preferences.
- Improved Operational Efficiency: Dynamic scheduling and route optimization reduce operational costs.
- Sustainable Urban Transportation: The system contributes to smart city initiatives by

optimizing bus routes, reducing fuel consumption, and lowering carbon emissions.

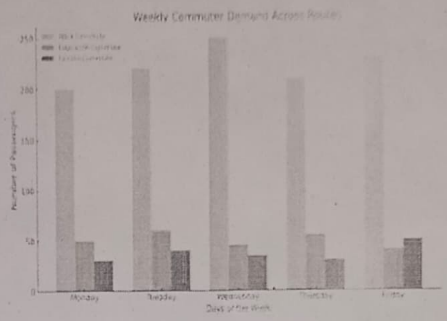

Fig 1.1

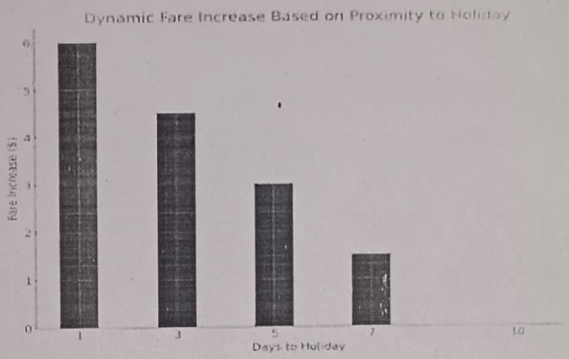

Fig 1.2

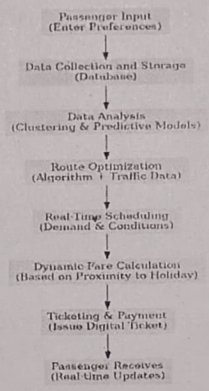

Fig 1.3

Future Enhancements:

The **Bus Onboarding System with Passenger Preference** has the potential for significant growth, both in terms of passenger experience and operational

efficiency. Future enhancements can evolve the system from a functional model to a comprehensive smart transportation solution capable of adapting to the changing needs of passengers, urban environments, and sustainability goals.

i. Integration with Multi- Modal Transport Systems
ii. Personalized Travel Recommendations and Dynamic Route Adjustments
iii. Dynamic Fare System Based on Demand and Proximity to Holidays
iv. AI-Powered Traffic Management and Real-Time Schedule Adjustments
v. Seamless Integration with Smart City Infrastructure
vi. Sustainability and Green Transport Initiatives
vii. Real-Time Passenger Feedback and Analytics

7. Conclusion:

This system represents a significant step toward transforming urban public transport, balancing efficiency with passenger-centric features. The dynamic fare model helps optimize demand management, while real- time route optimization ensures service reliability. Future work includes scaling the system for larger cities and integrating with broader smart city technologies for seamless urban mobility.

8. References:

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4.Chen, X., & Li, Y. (2019) "*Predictive analytics for passenger demand in public transportation systems" in Transportation Research Part B: Methodological*, 126, 213-225.

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C. SOURCE CODE

Manage.py

```
#!/usr/bin/env python
"""Django's command-line utility for administrative tasks."""
import os
import sys

def main():
    """Run administrative tasks."""
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'btrs_django.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
    execute_from_command_line(sys.argv)

if __name__ == '__main__':
    main()
```

Intial.py

```
# Generated by Django 4.0.3 on 2022-03-24 02:13

from django.db import migrations, models
import django.db.models.deletion
import django.utils.timezone

class Migration(migrations.Migration):

    initial = True

    dependencies = [
    ]

    operations = [
```

```
to='reservationApp.bus')),
                ('depart', models.ForeignKey(on_delete=django.db.models.deletion.CASCADE,
related_name='depart_location', to='reservationApp.location')),
                ('destination', models.ForeignKey(on_delete=django.db.models.deletion.CASCADE,
related_name='destination', to='reservationApp.location')),
            ],
        ),
    ]
```

Bus_Category.py

```
# Generated by Django 4.0.3 on 2022-03-24 02:54

from django.db import migrations, models
import django.db.models.deletion

class Migration(migrations.Migration):

    dependencies = [
        ('reservationApp', '0001_initial'),
    ]

    operations = [
        migrations.AddField(
            model_name='bus',
            name='category',
                                                    field=models.ForeignKey(blank=True,          null=True,
on_delete=django.db.models.deletion.CASCADE, to='reservationApp.category'),
        ),
    ]
```

Bus_Seat.py

```
# Generated by Django 4.0.3 on 2022-03-24 05:24

from django.db import migrations, models

class Migration(migrations.Migration):

    dependencies = [
        ('reservationApp', '0002_bus_category'),
    ]

    operations = [
```

```
        migrations.AddField(
            model_name='bus',
            name='seats',
            field=models.FloatField(default=0, max_length=5),
        ),
    ]
```

Booking.py

```
# Generated by Django 4.0.3 on 2022-03-24 05:51

from django.db import migrations, models
import django.db.models.deletion
import django.utils.timezone

class Migration(migrations.Migration):

    dependencies = [
        ('reservationApp', '0003_bus_seats'),
    ]

    operations = [
        migrations.CreateModel(
            name='Booking',
            fields=[
                        ('id', models.BigAutoField(auto_created=True, primary_key=True, serialize=False,
verbose_name='ID')),
                ('code', models.CharField(max_length=100)),
                ('name', models.CharField(max_length=250)),
                ('seats', models.IntegerField(max_length=11)),
                ('status', models.CharField(choices=[('1', 'Pending'), ('2', 'Paid')], max_length=2)),
                ('date_created', models.DateTimeField(default=django.utils.timezone.now)),
                ('date_updated', models.DateTimeField(auto_now=True)),
                            ('schedule', models.ForeignKey(on_delete=django.db.models.deletion.CASCADE,
to='reservationApp.schedule')),
            ],
        ),
    ]
```

Admin.py

```
from django.contrib import admin
from reservationApp.models import Category, Location, Bus, Schedule, Booking
# Register your models here.
admin.site.register(Category)
admin.site.register(Location)
admin.site.register(Bus)
admin.site.register(Schedule)
admin.site.register(Booking)
```

Apps.py

```
from django.apps import AppConfig

class ReservationAppConfig(AppConfig):
    default_auto_field = 'django.db.models.BigAutoField'
    name = 'reservationApp'
```

Forms.py

```
import sched
from unicodedata import category
from django import forms
from django.contrib.auth.forms import UserCreationForm,PasswordChangeForm, UserChangeForm

from django.contrib.auth.models import User
from more_itertools import quantify
from .models import Category, Location, Bus, Schedule, Booking
from datetime import datetime

class UserRegistration(UserCreationForm):
    email = forms.EmailField(max_length=250,help_text="The email field is required.")
    first_name = forms.CharField(max_length=250,help_text="The First Name field is required.")
    last_name = forms.CharField(max_length=250,help_text="The Last Name field is required.")

    class Meta:
        model = User
        fields = ('email', 'username', 'password1', 'password2', 'first_name', 'last_name')

    def clean_email(self):
        email = self.cleaned_data['email']
```

```
        try:
            user = User.objects.get(email = email)
        except Exception as e:
            return email
        raise forms.ValidationError(f"The {user.email} mail is already exists/taken")

    def clean_username(self):
        username = self.cleaned_data['username']
        try:
            user = User.objects.get(username = username)
        except Exception as e:
            return username
        raise forms.ValidationError(f"The {user.username} mail is already exists/taken")

class UpdateProfile(UserChangeForm):
    username = forms.CharField(max_length=250,help_text="The Username field is required.")
    email = forms.EmailField(max_length=250,help_text="The Email field is required.")
    first_name = forms.CharField(max_length=250,help_text="The First Name field is required.")
    last_name = forms.CharField(max_length=250,help_text="The Last Name field is required.")
    current_password = forms.CharField(max_length=250)

    class Meta:
        model = User
        fields = ('email', 'username','first_name', 'last_name')

    def clean_current_password(self):
        if not self.instance.check_password(self.cleaned_data['current_password']):
            raise forms.ValidationError(f"Password is Incorrect")

    def clean_email(self):
        email = self.cleaned_data['email']
        try:
            user = User.objects.exclude(id=self.cleaned_data['id']).get(email = email)
        except Exception as e:
            return email
        raise forms.ValidationError(f"The {user.email} mail is already exists/taken")

    def clean_username(self):
        username = self.cleaned_data['username']
        try:
            user = User.objects.exclude(id=self.cleaned_data['id']).get(username = username)
        except Exception as e:
            return username
        raise forms.ValidationError(f"The {user.username} mail is already exists/taken")

class UpdatePasswords(PasswordChangeForm):
        old_password = forms.CharField(widget=forms.PasswordInput(attrs={'class':'form-control form-
```

```
control-sm rounded-0'}), label="Old Password")
    new_password1 = forms.CharField(widget=forms.PasswordInput(attrs={'class':'form-control form-control-sm rounded-0'}), label="New Password")
    new_password2 = forms.CharField(widget=forms.PasswordInput(attrs={'class':'form-control form-control-sm rounded-0'}), label="Confirm New Password")
  class Meta:
    model = User
    fields = ('old_password','new_password1', 'new_password2')

class SaveCategory(forms.ModelForm):
  name = forms.CharField(max_length="250")
  description = forms.Textarea()
  status = forms.ChoiceField(choices=[('1','Active'),('2','Inactive')])

  class Meta:
    model = Category
    fields = ('name','description','status')

  def clean_name(self):
    id = self.instance.id if self.instance.id else 0
    name = self.cleaned_data['name']
    # print(int(id) > 0)
    # raise forms.ValidationError(f"{name} Category Already Exists.")
    try:
      if int(id) > 0:
        category = Category.objects.exclude(id=id).get(name = name)
      else:
        category = Category.objects.get(name = name)
    except:
      return name
      # raise forms.ValidationError(f"{name} Category Already Exists.")
    raise forms.ValidationError(f"{name} Category Already Exists.")

class SaveLocation(forms.ModelForm):
  location = forms.CharField(max_length="250")
  status = forms.ChoiceField(choices=[('1','Active'),('2','Inactive')])

  class Meta:
    model = Location
    fields = ('location','status')

  def clean_location(self):
    id = self.instance.id if self.instance.id else 0
    location = self.cleaned_data['location']
    # print(int(id) > 0)
    try:
      if int(id) > 0:
```

```
    status = forms.ChoiceField(choices=[('1','Active'),('2','Cancelled')])
        try:
            location = Location.objects.get(id=location_id)
            return location
        except:
            raise forms.ValidationError("Destination is not recognized.")

class SaveBooking(forms.ModelForm):
    code = forms.CharField(max_length="250")
    schedule = forms.CharField(max_length="250")
    name = forms.CharField(max_length="250")
    seats = forms.CharField(max_length="250")

    class Meta:
        model = Booking
        fields = ('code','schedule','name','seats')

    def clean_code(self):
        id = self.instance.id if self.instance.id else 0
        if id > 0:
            try:
                booking = Booking.objects.get(id = id)
                return booking.code
            except:
                code= ''
        else:
            code= ''
        pref = datetime.today().strftime('%Y%m%d')
        code = str(1).zfill(4)
        while True:
            sched = Booking.objects.filter(code=str(pref + code)).count()
            if sched > 0:
                code = str(int(code) + 1).zfill(4)
            else:
                code = str(pref + code)
                break
        print(code)
        return code

    def clean_schedule(self):
        schedule_id = self.cleaned_data['schedule']
        # print(int(id) > 0)
        try:
            sched = Schedule.objects.get(id = schedule_id)
            return sched
        except:
            raise forms.ValidationError(f"Trip Schedule is not recognized.")
```

```
class PayBooked(forms.ModelForm):
    status = forms.IntegerField()

    class Meta:
        model = Booking
```

Models.py

```
from re import I
from unicodedata import category
from django.db import models
from django.utils import timezone
from django.dispatch import receiver
from more_itertools import quantify
from django.db.models import Sum

# Create your models here.

class Category(models.Model):
    name = models.CharField(max_length=250)
    description = models.TextField()
    status = models.CharField(max_length=2, choices=(('1','Active'),('2','Inactive')), default=1)
    date_created = models.DateTimeField(default=timezone.now)
    date_updated = models.DateTimeField(auto_now=True)

    def __str__(self):
        return self.name

class Location(models.Model):
    location = models.CharField(max_length=250)
    status = models.CharField(max_length=2, choices=(('1','Active'),('2','Inactive')), default=1)
    date_created = models.DateTimeField(default=timezone.now)
    date_updated = models.DateTimeField(auto_now=True)

    def __str__(self):
        return self.location

class Bus(models.Model):
    category = models.ForeignKey(Category,on_delete=models.CASCADE, blank= True, null = True)
    bus_number = models.CharField(max_length=250)
    seats = models.FloatField(max_length=5, default=0)
    status = models.CharField(max_length=2, choices=(('1','Active'),('2','Inactive')), default=1)
    date_created = models.DateTimeField(default=timezone.now)
    date_updated = models.DateTimeField(auto_now=True)
```

```
    def __str__(self):
        return self.bus_number

class Schedule(models.Model):
    code = models.CharField(max_length=100)
    bus = models.ForeignKey(Bus,on_delete=models.CASCADE)
    depart = models.ForeignKey(Location,on_delete=models.CASCADE, related_name='depart_location')
    destination = models.ForeignKey(Location,on_delete=models.CASCADE, related_name='destination')
    schedule= models.DateTimeField()
    fare= models.FloatField()
    status = models.CharField(max_length=2, choices=(('1','Active'),('2','Cancelled')), default=1)
    date_created = models.DateTimeField(default=timezone.now)
    date_updated = models.DateTimeField(auto_now=True)

    def __str__(self):
        return str(self.code + ' - ' + self.bus.bus_number)

    def count_available(self):
        booked = Booking.objects.filter(schedule=self).aggregate(Sum('seats'))['seats__sum']
        return self.bus.seats - booked

class Booking(models.Model):
    code = models.CharField(max_length=100)
    name = models.CharField(max_length=250)
    schedule = models.ForeignKey(Schedule,on_delete=models.CASCADE)
    seats = models.IntegerField()
    status = models.CharField(max_length=2, choices=(('1','Pending'),('2','Paid')), default=1)
    date_created = models.DateTimeField(default=timezone.now)
    date_updated = models.DateTimeField(auto_now=True)

    def __str__(self):
        return str(self.code + ' - ' + self.name)

    def total_payable(self):
        return self.seats * self.schedule.fare
```

Test.py

```
from django.test import TestCase

# Create your tests here.
```